OPEN PLATFORM FIRMWARE (OPF)



From Code to Data: Advancing the Unified Platform Configuration Interface (UPCI)

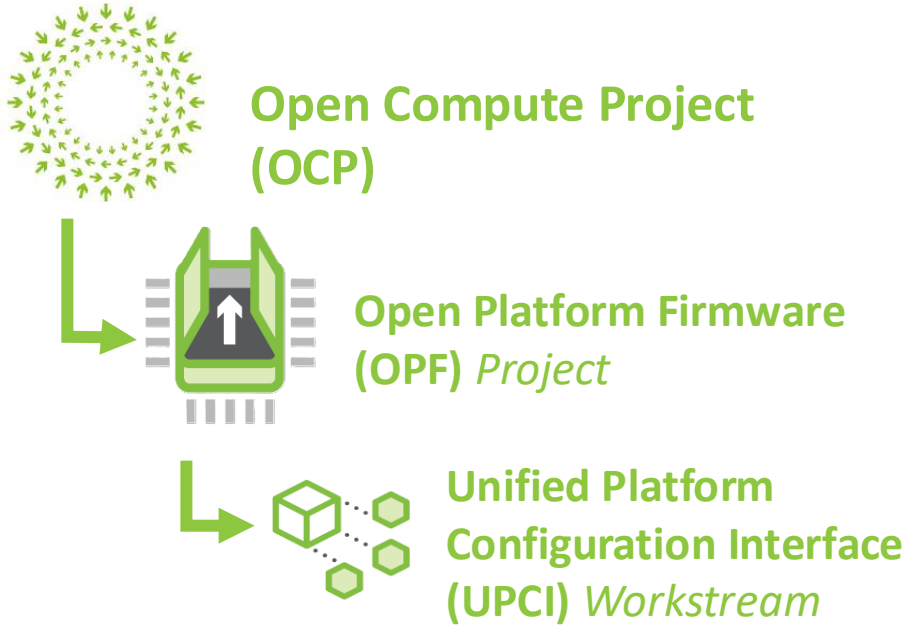
Felix Polyudov *Fellow, AMI*

Tim Lewis *CTO, Insyde Software*

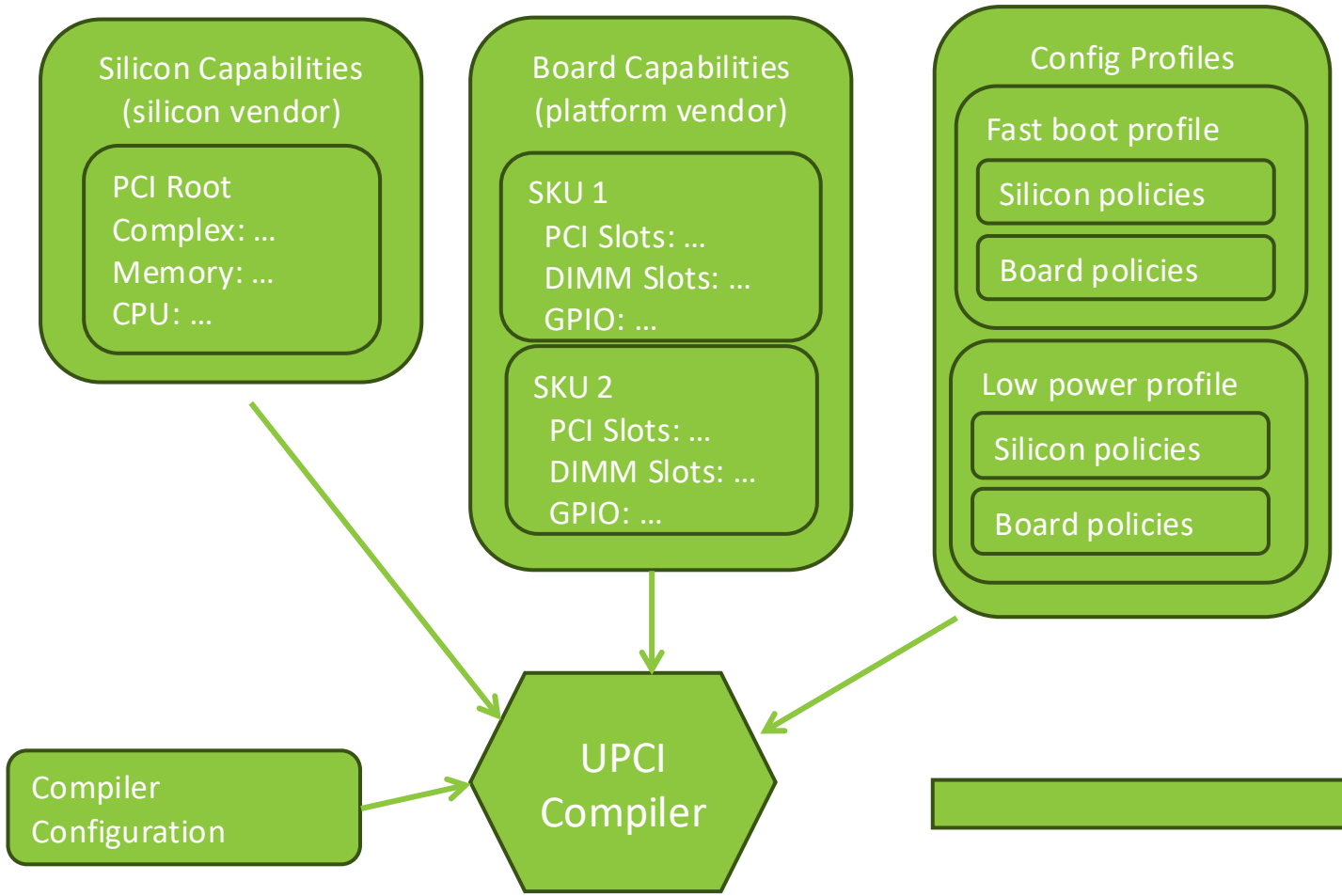


UPCI Recap

- UPCI is a workstream of the Open Platform Firmware Project focused on establishing open, standardized platform configuration mechanisms
- Aims to shift platform configuration from a code-driven to a data-driven model
- Keys activities
  - Human Readable Representation (HRR)
    - Development time configuration format
  - Machine Readable Representation (MRR)
    - Execution time configuration format
  - Mapping to other industry standards



Development Time Architecture



- UPCI compiler receives input in the HRR format
- It produces config data in requested output formats (based on compiler configuration)
- One of the output types is schema that can be used for runtime conversion between formats

Human Readable Representation

- YAML and JSON notation
- Separate format and data specifications
  - Format specification has higher complexity and is defined by the UPCI
  - Data definition are defined by the Silicon/Platform/Firmware vendor based on UPCI format specification
- Extensible
  - Future versions of the specification may add new data structures or extend existing data structures
  - Data structures are versioned
- Required and Optional Fields
  - Some fields may be omitted during output generation based on platform configuration profile

HRR Example (format and data descriptors)

Format Description (created by UPCI)

```
ADDRESS_RANGE:
  version: 1
  description: "Address range entry"
  fields:
    - name: start
      type: u64
      description: "Start address of the range"
    - name: size
      type: u64
      description: "Size of the range in bytes"
    - name: range_type
      type: u32
      constraints:
        kind: enum
        values: [0 : “reserved”, 1 : “memory”]
      description: "Type of the address range"
```

Data Definition (created by firmware owner)

```
Range:
  format: ADDRESS_DESCRIPTOR
  start: 0x0000000000000000
  size: 0x0000000000100000
  type: memory
```

HRR Example (extend/override capabilities)

Format Description (created by UPCI)

```
PCIE_BIFURCATION16:
  version: 1
  prompt:
      en-US: Bifurcation mode
  fields:
    - name: mode
      type: u8
      default: auto
      constraints:
        kind: enum
        values: [0 : “auto”, 1 : “x16”, 2:” x8x8”, …]
```

Baseline Defintion(created by Silicon Vendor)

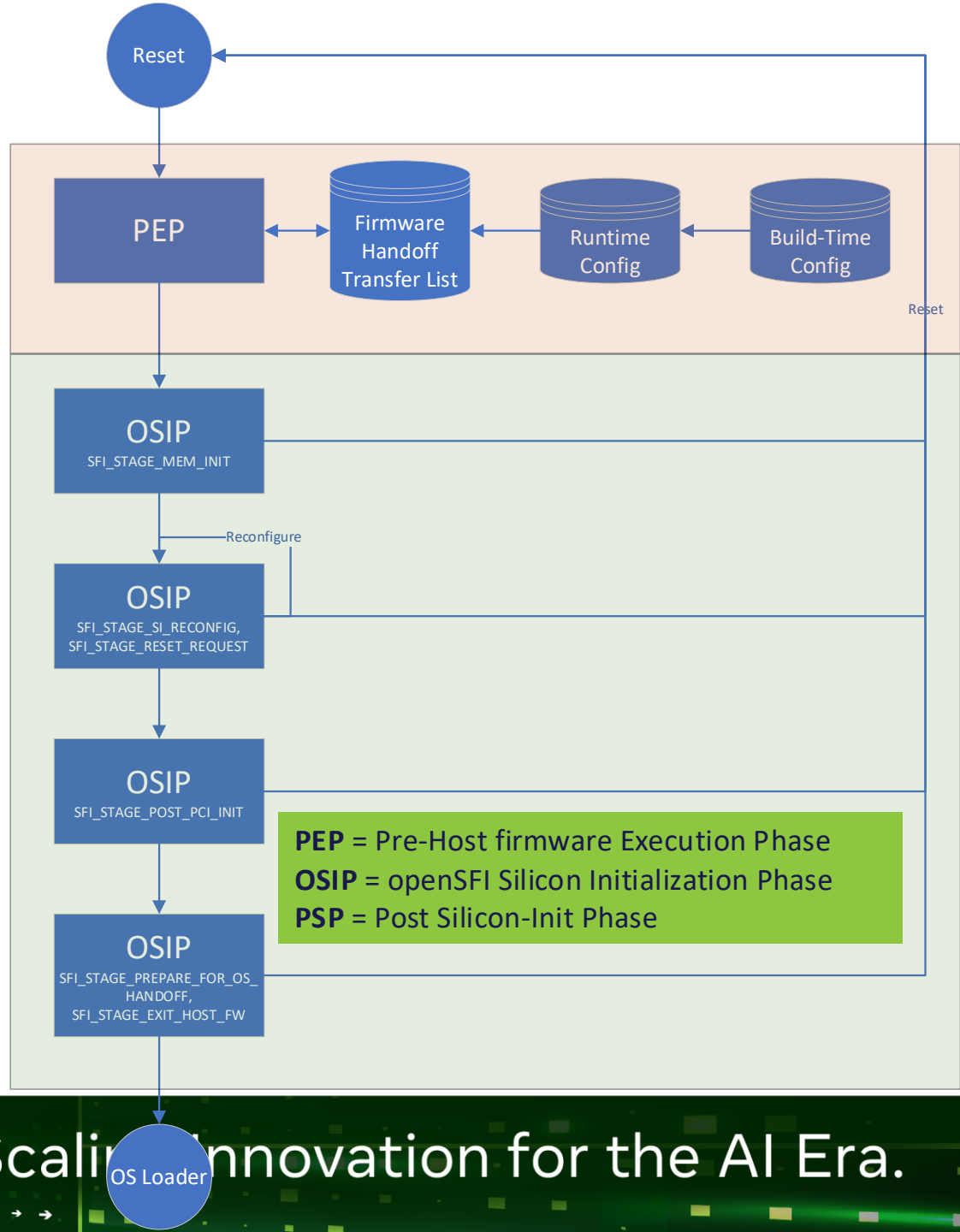
```
PciBifurcation16:
  format: PCIE_BIFURCATION
  mode: auto
```

Customization (created by Platform Vendor)

```
PciBifurcation16:
  format: PCIE_BIFURCATION
  mode: x16
  prompt:
      fr-FR: Mode de bifurcation
      de-DE: Bifurkationsmodus
      zh-chs: 拆分模式
```

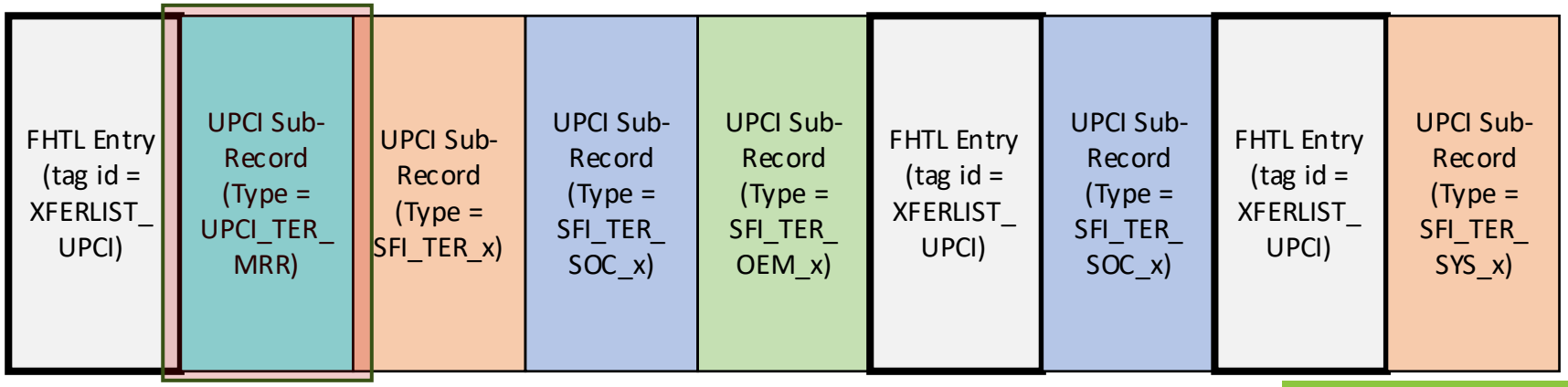
UPCI Firmware Handoff Transfer List (FHTL):

- Build tools create the build time FHTL.
- PEP reads the build-time FHTL and integrates the updatable FHTL to create the runtime FHTL.
- PEP uses the runtime FHTL to perform SoC initialization.
- PEP passes the runtime FHTL to the Host Firmware entry point.
- Host Firmware maintains the runtime FHTL during the OS Initialization process.
- Portions of the FHTL may be marked for non-volatile storage for use during the next boot.



HRR ➔ MRR ➔ FHTL

- The UPCI compiler converts HRR to a single TE of type **`XFERLIST_UPCI`** at build time, selecting the MRR for one or more SKUs.
- Additional XFERLIST_UPCI TEs and TE Records may be created OR existing TE Records may be marked empty (to delete them) or updated.
  - 0x00000000 – Record list terminator (**`SFI_TER_END`**)*
  - 0x00000001 – GUID-defined (**`SFI_TER_GUID`**)*
  - 0x00000002 – UPCI MRR (**`UPCI_TER_MRR`**)*
  - 0x000001xxx – SFI (SFI_TER)
  - 0x00000exxx – Reserved for SoC vendors (**`SFI_TER_SOC`**)*
  - 0x00000fxxx – Reserved for OEM usage (**`SFI_TER_OEM`**)*



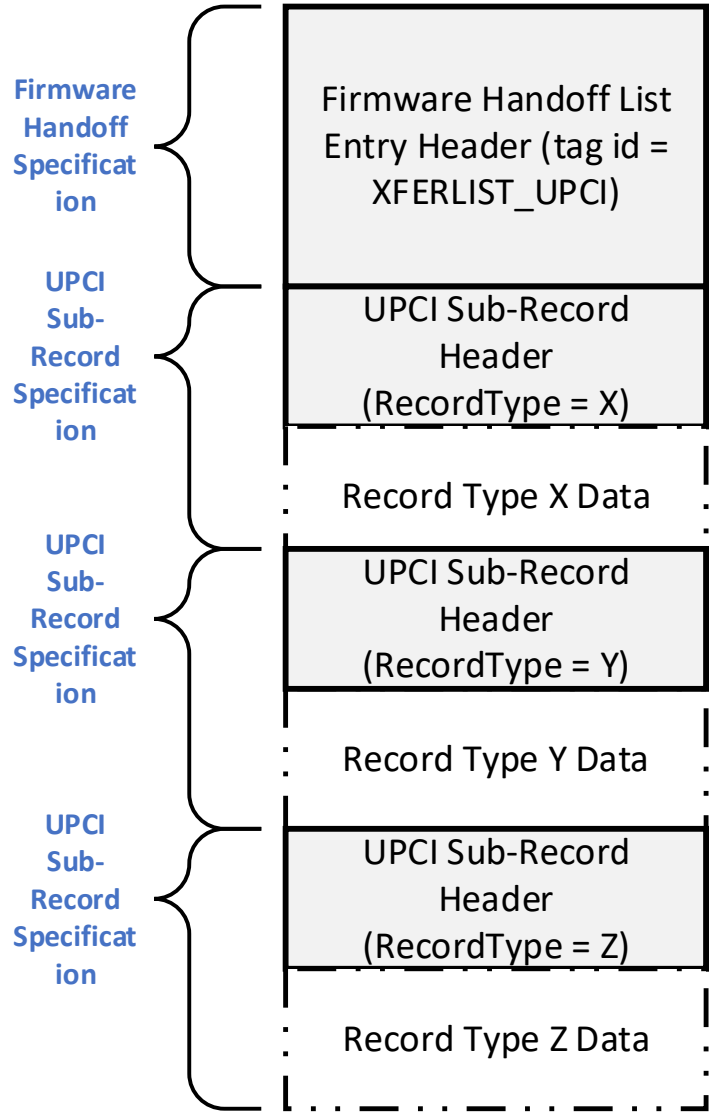
*Proposed record types

Build FHTL (bFHTL)

- Build-time Firmware Handoff Transfer Lists (bFHTL) and Updatable Firmware Handoff Transfer Lists (uFHTL) are created by the build tools when creating the firmware image.
- The bFHTL and uFHTL are read and validated by the PEP code using a SoC-vendor-specific method in the earliest phases of the boot process and merged.

Runtime FHTL (rFHTL)

- The Runtime Firmware Handoff Transfer List (rFHTL) is a zero-initialized buffer provided to the host firmware entry point by PEP and must contain the Firmware Handoff header and zero or more Transfer List Entries (TEs).
  - The maximum buffer size and current used size is indicated in the Firmware Handoff header
- UPCI and openSFI use a single TE type with the tag_id **`XFERLIST_UPCI`***.
  - There may be more than one UPCI TE in the rFHTL.
- TE types from the Firmware Handoff specification may be used but their usage may be constrained by the UPCI or openSFI specification.
  - Ex: The **`XFERLIST_ACPI_AGGR`** and **`XFERLIST_HOB_`**L are defined as transferring an entire table, but for UPCI and openSFI usage these will hold only a single record.
- Each UPCI TE contains zero or more Transfer list Entry Records (TER) with an enumerated type.
  - TER types are defined in the UPCI specification:
- TERs which are designed to aid in the creation of a multi-part runtime table (i.e. ACPI SRAT, ACPI MCFG) will provide the smallest logical sub-record type rather than the entire table.



* Awaiting official allocation from Firmware Handoff team.

Updatable FHTL (uFHTL)

- Updatable FHTL are entries designed to be modified during the boot process and accessed on subsequent boots.
- Default entries for the FHTL may be provided during the build time.
- Read and merged into the runtime FHTL by the PEP phase.
- Marked as “non-volatile” in the flags.
- May be written to non-volatile storage in one or more openSFI stages.
- Format is specified in the openSFI specification.

Call to Action

- Problems to solve
  - Finalize HRR/MRR schemas
  - Define UPCI data structures for hardware sub-domains (CPU, Memory, PCI-E, etc.)
- How to get involved in the project/sub-project community
  - Join OPF Mailing list: https://ocp-all.groups.io/g/OCP-OPF
  - Join OPF by-weekly calls: https://ocp-all.groups.io/g/OCP-OPF/calendar
- Where to find additional information
  - OPF Wiki: https://github.com/opencomputeproject/OpenSystemFirmware/wiki
  - UPCI Wiki: https://github.com/opencomputeproject/OpenSystemFirmware/wiki/Unified-Platform-Configuration

Thank You!

